CHAPTER 7

AMONG THE MOONS OF SATURN

Seat thyself sultanically among the moons of Saturn.

—HERMAN MELVILLE, *MOBY DICK*, CHAPTER 107 (1851)

There is a world, midway in size between the Moon and Mars, where the upper air is rippling with electricity—streaming in from the archetypical ringed planet next door, where the perpetual brown overcast is tinged with an odd burnt orange, and where the very stuff of life falls out of the skies onto the unknown surface below. It is so far away that light takes more than an hour to get there from the Sun. Spacecraft take years. Much about it is still a mystery—including whether it holds great oceans. We know just enough, though, to recognize that within reach may be a place where certain processes are today working themselves out that aeons ago on Earth led to the origin of life.

On our own world a long-standing—and in some respects quite promising—experiment has been under way on the evolution of matter. The oldest known fossils are about 3.6 billion years old. Of course, the origin of life had to have happened well before that. But 4.2 or 4.3 billion years ago the Earth was being so ravaged by the final stages of its formation that life could not yet have come into being: Massive collisions were melting the surface, turning the oceans into steam and driving any atmosphere that had